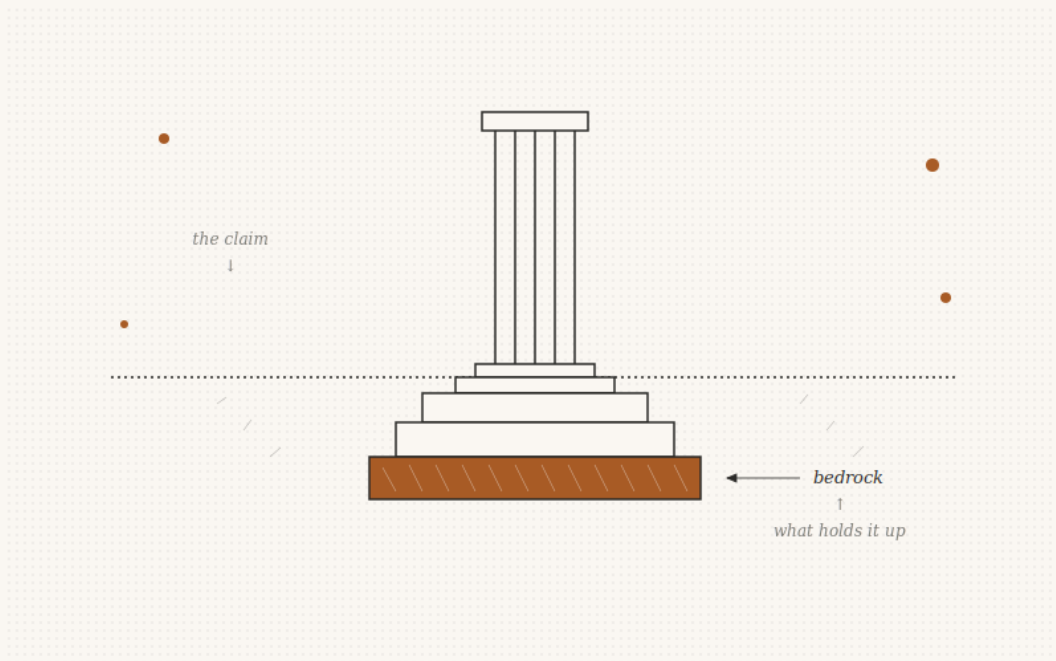


A LECTURE FROM COAST

First-Principles Thinking

reasoning from the bedrock, not the analogy



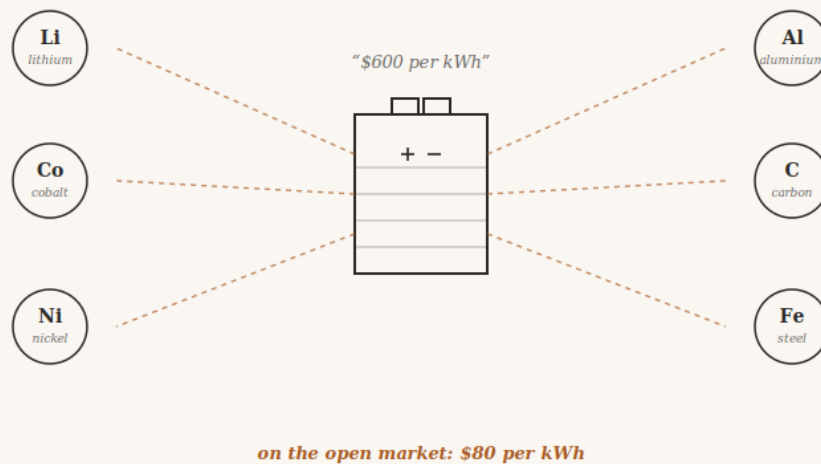
Most of what you believe, you inherited. You did not test it; you absorbed it. There is an older way of thinking — Aristotle named it, Descartes refined it, and the people who actually reshape the world still use it.

A young engineer in 2003

In 2003, a thirty-one-year-old engineer named Elon Musk wanted to build an electric car company. Everyone he spoke to gave him the same warning: the batteries make it impossible. Batteries cost \$600 per kilowatt-hour. They had been at roughly that price for years. They would always be at roughly that price, because that was what batteries cost.

Musk did something unusual. He asked: *what is a battery made of?*

A lithium-ion battery is made of cobalt, nickel, aluminium, carbon, polymer separators, and a steel can. He went to the London Metal Exchange and looked up the price of each material — not the price of finished batteries, but the price of the raw materials, on the open market, in the quantities required.



The same battery, decomposed into the open-market price of its raw materials.

The difference between \$80 and \$600 wasn't physics. It was a story — a story the battery industry told itself about what batteries cost, repeated until it became 'what everyone knows'. The story was real in the sense that it described the current market. It was not real in the sense of being a law.

That gap, between what is and what must be, is where first-principles thinking lives.

Module 2

An ancient word

The technique is older than electric cars by 2,300 years.

In *Posterior Analytics*, Aristotle defined a *first principle* — in Greek, ἀρχή, *arkhē* — as ‘the first basis from which a thing is known.’ Not borrowed. Not inherited. Not assumed because everyone else assumes it. The bedrock from which everything else can be reasoned.

Aristotle's point was simple and uncomfortable: most of what we believe is not built on first principles. It is built on analogy. *This is true because it resembles that other thing which is true.* Analogy is fast, and it usually works, and it is how human cognition mostly runs. But when the analogy is wrong — when the thing you're reasoning about doesn't actually resemble the thing you're comparing it to — analogy will quietly carry you off a cliff.

First-principles thinking is the antidote. It is slow, it is rare, and it is the technique behind almost every moment in history where someone did something that other people said was impossible.

This lesson teaches it.



Module 3

The two modes

There are two ways to know something.

The first is *by analogy*. You believe X because X is like Y, and Y is true. You believe a battery costs \$600 because batteries have always cost \$600. You believe a startup needs venture funding because that's how startups work. You believe a degree leads to a job because it has, historically, for people like you.

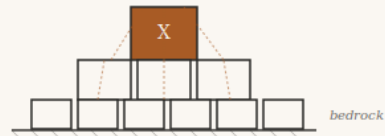
The second is *by first principles*. You believe X because you have decomposed X to its irreducible parts, examined each part, and rebuilt X from those parts. The battery costs what its materials cost. The startup needs whatever capital its actual operations require, no more. The degree provides whatever its actual content does, no more.

by analogy



*a chain of resemblances —
fast, social, often right
...sometimes wrong*

from first principles



*built up from irreducibles —
slow, lonely, right where it matters*

The same conclusion (X), arrived at by two completely different routes.

Reasoning by analogy is the default of the human mind. It is fast, social, and right often enough to survive. Reasoning from first principles is effortful, lonely, and right in places where analogy fails — which happens to be the places that matter most.

The goal of this lesson is not to make you reason from first principles always. That would be exhausting and unnecessary. The goal is to recognise *which* of your beliefs are inherited, and to give you the tool to take them apart when it matters.

The procedure

The procedure is three steps.

- 1. Take a belief or claim that matters to you, and write it down.** Be precise. ‘Crypto is a scam’ is not precise. ‘Buying Bitcoin in 2026 is a bad investment for me’ is precise.
- 2. Decompose it.** Ask: *what are the foundational facts on which this rests?* Keep asking *why is that true?* until you hit something you can't reduce further — either a physical fact, a measurable quantity, or an empirical observation. Write each layer down.
- 3. Reconstruct.** Now build the claim back up, but using only the irreducible facts. See what survives. Sometimes the claim survives intact. Sometimes it survives with revisions. And sometimes, like Musk's battery, it does not survive at all.

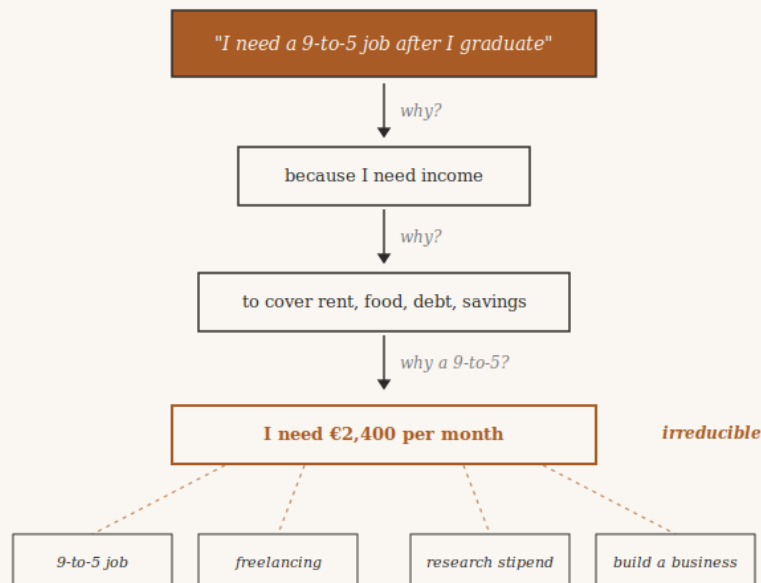
That is the entire method. Most of the work is in step 2. Most of the courage is in step 3.

A worked example

Take this belief: *I need to get a 9-to-5 job after I graduate.*

Step 1 — write it down precisely. ‘After my degree, I need to take a full-time salaried job because that is what people do.’

Step 2 — decompose. Why? *Because I need income.* Why income? *To pay rent, food, healthcare, debt, leisure, savings.* Why does income require a 9-to-5? *Because that is the standard arrangement.* Why is that the standard? *Because employers pay reliable salaries in exchange for predictable hours.* But is that the only way to get income? *No — freelancing, contracts, starting a business, savings, family support, scholarships, deferred earnings.* What is the actual irreducible requirement? *That I have enough money each month to cover what I need.* What is that number? *Calculate it. Let's say it's €2,400 per month including buffer.*



Each 'why' strips a layer until you reach the irreducible requirement underneath.

Step 3 — reconstruct. The actual claim becomes: 'I need to generate €2,400 per month, somehow, after I graduate.' A 9-to-5 is one way to do this. It is not the only way. Some of the alternatives — freelancing in a high-rate field, working part-time and building something on the side, taking a research stipend — may fit my life better. The original belief was an inherited assumption smuggling itself in as a necessity.

This does not mean you should not take a 9-to-5. It means that if you do, you should do so because you chose to, not because you didn't realise there were other options.

That is what first principles does. It does not always change the answer. It always changes whether you can be said to have actually chosen it.

Module 6

Where the technique fails

It is worth being honest about the limits.

First-principles thinking is expensive. It takes hours per belief, sometimes days. You cannot apply it to everything. If you tried, you would never make a decision again.

It also fails when you do not have access to the underlying facts. Musk could check the London Metal Exchange. If he had been trying to first-principle his way to whether God exists, the technique would not have helped, because the underlying facts are not available.

And it can be self-deceptive. People sometimes claim to be reasoning from first principles when they are really reasoning from a different analogy than the mainstream one — replacing one inherited story with another, more rebellious-sounding one. The test is: can you point to the irreducible facts? If yes, it is first principles. If no, it is just contrarianism.

Use the technique on the beliefs that *matter most* — career, money, health, where to live, what to study, what to build — and accept analogy for the rest.

Module 7

A famous gallery

The history of breakthroughs is largely a history of first-principles thinking.

a brief gallery of refusal



Galileo was told that heavy objects fell faster than light ones, because Aristotle had said so and everyone had repeated it for 1,900 years. He went to the Leaning Tower of Pisa and dropped two balls. They hit the ground at the same time. He had not invented physics. He had merely refused to inherit it.

The Wright brothers were told that powered flight was impossible. They did not believe powered flight was easy. They believed that *what they had been told about why it was impossible* was based on analogy to birds, not on actual aerodynamics. They built a wind tunnel and tested airfoils. They reasoned from lift and drag, not from what other people had failed to do.

Marie Curie was told that radioactivity was a property of uranium. She asked: *is it?* She measured pitchblende — the ore — and found it radioactive far beyond what its uranium content predicted. There must be another element. She isolated polonium. Then radium.

In every case, the pattern is the same. Someone refuses to inherit a belief. They go down to the irreducible facts. They rebuild.



Module 8

Exercise — apply the technique

Take a belief you hold strongly. Something where, if challenged, you would say *of course X* with confidence. Examples: *Of course you need a university degree.* *Of course you can't make money doing what you love.* *Of course remote work is worse.* *Of course this country is in decline.* *Of course AI will or will not take my job.*

Pick one. Write it down precisely. Now run the procedure:

1. What does this claim depend on? List the underlying assumptions.
2. For each assumption, ask: how do I know this is true? Keep going until you reach something irreducible — a physical fact, a measurement, an actual experience.
3. Rebuild the claim using only the irreducibles. Compare to the original.

Do this for one belief, fully. It will take you fifteen minutes. You will end the exercise either more confident in the belief (because it survived) or holding a more accurate, more defensible version of it. Either outcome is a win.

Module 9

A subtle point

There is a famous misuse of first-principles thinking that's worth flagging.

People sometimes use the phrase to mean *I ignore what experts say and figure it out myself*. That is not first-principles thinking. That is *arrogance*. Real first-principles thinking respects expertise — experts often have access to irreducible facts that you don't — but refuses to confuse expertise with received wisdom.

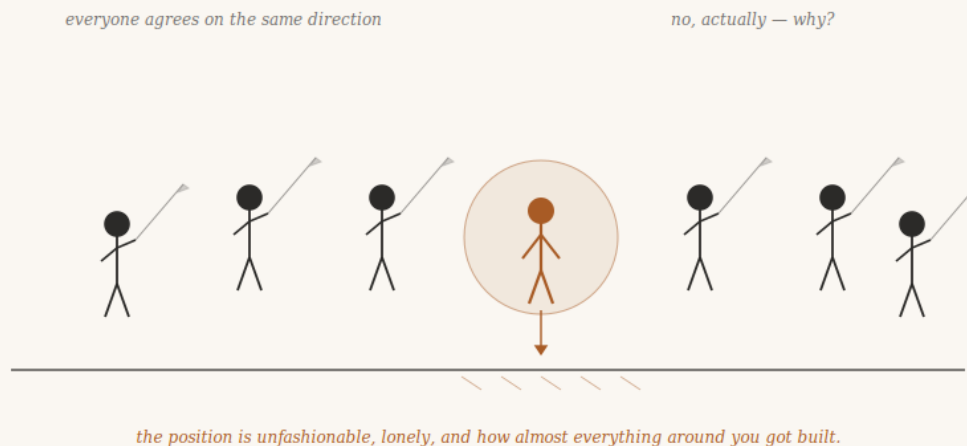
The question to ask an expert is not *what do you think?* It is *what do you actually know, and how do you know it?* The first question yields opinion. The second yields data. A good expert can distinguish these. A bad one cannot. If they cannot, you have learned something useful.

Use experts as sources of facts, not sources of conclusions. The conclusions are yours to draw.

Module 10

A closing thought

There is a particular feeling that comes with first-principles thinking, and it is uncomfortable.



It is the feeling of standing alone in a room where everyone else has agreed on something, and saying *no, actually, why?* It is unfashionable. It is socially expensive. It is what gets you called naive, or arrogant, or both.

It is also what built almost everything around you. The houses, the medicine, the laws, the institutions, the technologies — none of them existed because someone inherited them. They existed because someone, at some point, refused to.

You are not obligated to live a first-principles life. Most people don't, and most people are fine. But on the few questions in your life that genuinely matter — what to do, what to build, what to believe, who to be — the technique is available to you. It costs an evening per question. It will sometimes save you years.

Welcome to one of the oldest tools in human thought.



What to learn next on Coast

- ◆ *The Polya Method* — four steps that turn lunging into thinking.
- ◆ *Fermi Estimation* — irreducible facts in the service of order-of-magnitude answers.
- ◆ *The Socratic Method* — five questions that disassemble any belief.